BATCH 9 · FOR THE TEAM ONLY — DO NOT SUBMIT THIS FILE

Presentation Script and Q&A Pack

Credit Pre-Screening Engine — MUC02 · 17 slides · 22–24 minutes · Four speakers

How to use this. Each slide has one job and 2–4 sentences to say. You do not have to say them word for word — say them in your own words, but keep the **numbers in bold** exactly as they are. Those numbers are what the marks are for.

Who speaks when

Speaker	Slides	Time	Their part of the story
Mohammed Mutee Ruknuddin	1 – 6	~6 min	The bank, the problem, the canvas, how we worked, cleaning the data
Mohammed Saalif Udyawar	7 – 9	~5 min	Three findings: the 650 cliff, the FOIR cliff, gig workers
Mulla Javeed	10 – 12	~6 min	The model, how we judge it, and choosing the threshold
Nagothu Varshitha	13 – 17	~6 min	The blind spot, fairness, the money, the actions, the limits

Handover line — say the next person's name so the audience knows the section changed. For example: “Saalif will now show you what we found in the data.”

Mohammed Mutee Ruknuddin

Slide 1 — Title

Time: ~30 sec

- Good morning. We are Batch 9. Our use case is the **Credit Pre-Screening Engine** for a retail bank's personal loan business.
- The question we were given is on the screen: can six months of the bank's own decisions tell us which loan applications we never needed to read?
- I am Mutee, I did the data cleaning and the problem canvas. Saalif found the drivers, Javeed built the model, and Varshitha did fairness and the final briefing.

Slide 2 — A day inside the lending unit

Time: ~1 min

- Picture the bank. It gives personal loans across India — salaried people, business owners, gig workers, in big cities and small ones.
- Every month **3,000 applications** come in. Every single one is read by a human analyst, and that takes about **25 minutes**.
- Multiply those: **1,250 hours a month**. That is roughly eight people doing nothing but reading forms. And **47%** of what they read ends in a no.

Slide 3 — The problem is what fills the queue

Time: ~1 min

- Here is the real problem. It is not that there are too many applications. It is **what is in the queue**.
- Almost half get declined, and a lot of those are obvious in the **first minute** — the credit score is far too low, or the person's existing EMIs already eat most of their salary.
- But the analyst still spends the full 25 minutes on it. So the bank cannot grow without hiring, analysts spend their day on easy no's, and good customers wait in the queue behind files nobody needed to read.
- That is why the Head of Retail Lending asked the question in the yellow box.

Slide 4 — CBIM Problem Canvas

Time: ~1 min 15

- Before we touched the data, the four of us filled this canvas in together. It took 20 minutes and it changed how we worked.
- Situation and Complication you have just heard. The **Question** is the important one — notice it is not “build a model”. It is: which applications can leave the queue, at what confidence, without wrongly declining more than **5%** of customers we would have approved.

- And look at **SSOT**. The decision column is **what our credit team decided in the past** — not whether the customer actually repaid. That one sentence is the biggest limitation in our whole project, and we will come back to it at the end.

Slide 5 — Seven steps, four owners

Time: ~45 sec

- We worked in seven steps and split them four ways. Each step has code and a plain-English explanation of what it found.
- The order matters because each step feeds the next. My cleaning decides the features, the features decide the model, the model decides the threshold — and Varshitha's briefing has to defend all four.

Slide 6 — Cleaning: three problems, three decisions

Time: ~1 min 30

- My job was step one. Three things in this file needed a decision.
- **441 missing incomes.** They were not random — about **13% of business owners** and 11% of self-employed, but under 1% of salaried people. That makes sense: someone without a salary slip has no single clean number to write down. We filled them with the median income of **their own employment type**.
- **1,039 blank credit scores.** We checked — credit history was **zero months for every single one**. These are not errors. These are people who have never borrowed before. We kept them and flagged them.
- **12 impossible incomes**, over six lakh rupees a month, all salaried. Almost certainly annual income typed into a monthly box. We divided by 12, and every one landed back in a normal salary band.
- The point of the green box: we kept **all 12,000 rows**. One line of `dropna()` would have deleted 9% of applicants — the young first-time borrowers and the self-employed. Those are exactly the customers this bank is trying to grow.
- **Handover:** "Saalif will now show you what we found."

Mohammed Saalif Udyawar

Slide 7 — Finding 1 — the 650 cliff

Time: ~1 min 30

- My job was to find out what the credit team actually does — before anyone built a model.
- Look at this chart. Below a credit score of 650, approval is **3%** across **1,179** applications. Just above 650, it jumps to **32.5%**. At 800 plus, **85%**.
- That is a **tenfold jump across one single point** on the scale. That is not a gradient, it is a **rule**. Nobody has written "650" into the credit policy, but every analyst applies it.
- And see the orange bar — people with **no credit score at all** sit right in the middle at **53.9%**. There the team is using judgement, not a rule. Please remember that bar, because Varshitha comes back to it at the end.
- Why it matters: this is the **safest thing in the whole file to automate**. The machine is not being asked to judge. It is being asked to repeat a decision the team has already made the same way 1,179 times.

Slide 8 — Finding 2 — the FOIR cliff

Time: ~1 min 30

- This is the one I am most proud of, because this number **was not in the file. We built it.**
- FOIR is what Indian lenders use to measure affordability: the EMIs you already pay, plus the new EMI, divided by your monthly income. A FOIR of 0.6 means 60% of your salary goes to loan repayments.
- Up to 50%, approval falls gently — **65%, then 59%, then 47%**. Then it **collapses**: 47% to **32%** to **15.6%**. Approval more than halves at one line.
- So our recommendation is a written policy: **FOIR limit 50%**, **50 to 55% a referral band** that needs a reason, and **above 55% a decline** unless there is a documented exception.
- And notice — this is not a new rule we invented. It is the bank's own unwritten rule, made explicit, so every analyst applies it the same way.

Slide 9 — Finding 3 — gig workers

Time: ~1 min 15

- Third finding. Salaried applicants get approved **58.4%** of the time. Gig and contract workers, only **34.9%** — a 23-point gap.
- But — and this is the important part — **the job label is not the cause**. Gig applicants have a median income of **₹28,000 against ₹50,000** for salaried applicants. The same loan gives them a much higher FOIR.
- So employment type is really standing in for **income stability**, not for the job title.

- Why that is good news: the bank is declining on **affordability, not occupation** — which is defensible. And it points to a product fix. Offer a gig applicant a smaller loan or a longer tenure, their FOIR drops under 50%, and a decline becomes a sale.
- **Handover:** “Javeed will now take you through the model.”

Mulla Javeed

Slide 10 — The model

Time: ~1 min 45

- I built the model. It is a **logistic regression** with 12 features, trained on **9,600** applications with **2,400 held back** for testing.
- Two choices worth stating. The target is **Declined = 1**, because auto-declining is the action we want to take. And we left out **gender, city and the decision column** — gender because a bank simply cannot defend a credit decision that uses it.
- The chart shows what the model learned. Orange pushes towards decline, blue pulls towards approve. The top three are **FOIR at +1.22, credit score at −1.01, and missed payments at +0.98**.
- Now look at those three again. They are **exactly the three drivers Saalif found by hand, in the same order.**
- That is the whole point of the green box. **We never told the model about the 650 cliff or the 50% FOIR line. It found both on its own.** That is the strongest evidence we have that it learned the bank's real credit policy — not a fluke of this particular six months.

Slide 11 — How we judge it

Time: ~2 min

- Before I give you a result, let me explain how a model like this is judged, because the headline number is misleading.
- This is a **confusion matrix**. It is just a 2-by-2 table: what the model said, against what really happened.
- Top-left, **1,046** — model said approve, the team approved. Correct. Bottom-right, **852** — model said decline, the team declined. Also correct.
- Bottom-left, **277 false negatives** — the model said approve but the team declined. That is the **cheap** mistake. It costs 25 minutes of an analyst's time.
- Top-right is the one that matters. **225 false positives.** The model says decline, but the credit team had **approved** that person. That is a real customer, turned away by a machine.
- Two more words you will hear. **Precision 79.1%** — of everyone the model declines, how many really were declines. **Recall 75.5%** — of all the real declines, how many the model caught. Accuracy is also **79.1%**, and ROC-AUC is **0.867**.
- But read the red box. “The model is 79% accurate” describes the model. “**225 people we would have approved get turned away**” describes the business. The bank only cares about the second one.

Slide 12 — Finding 4 — the threshold

Time: ~1 min 45

- So here is the fix. A logistic regression does not give you a decision. It gives you a **probability**. Python's default cut-off is 0.5 — that is a statistical convention, it has nothing to do with lending.
- At 0.5, this model auto-declines **44.9%** of applications — but **one in five of those is someone the bank wanted**. No lending unit would sign that off.
- So the bank states its own limit instead, in the yellow box: **no more than 5%** of auto-declines may be people we would have approved.
- Now the model's only job is to show where that line falls. Follow the orange line down. At 0.85 we are at 5.9% — over the limit. At **0.90 we are at 4.7% — inside it.**
- So we recommend **0.90**. It removes **15.2% of applications: about 456 a month, and 190 analyst hours.**
- And note we take the **lowest** threshold that passes, not the safest one. Anything higher buys safety the bank has already said it does not need, and pays for it in lost volume.
- **Handover:** “Varshitha will tell you where this model should not be trusted.”

Nagothu Varshitha

Slide 13 — Finding 5 — the blind spot

Time: ~1 min 45

- Everything so far was the good news. Here is what we found when we looked underneath the average.
- The model is **81.2% accurate on people who have a credit score**, and only **56.7% on people who do not**. That is barely better than tossing a coin.
- And the reason is not statistics, it is **structure**. Think about it — someone who has never borrowed has no loans to be late on. So missed payments, active loans and existing EMIs are **all zero, by definition**, for all 1,039 of them. Three of the model's four strongest features are **blank** for this group. No amount of modelling fixes that.
- And it is already causing harm. At our 0.90 threshold, **12 auto-declines were new-to-credit people — and 6 of those 12 were applications the bank had approved.**
- So our fix is Design B: **never auto-decline someone with no credit history**. Send them to a human, always.
- And look what happens. Because we removed the group the model cannot read, we can lower the cut-off to **0.88** — and we get **more applications removed (16.4% vs 15.2%) AND fewer mistakes (4.1% vs 4.7%)** at the same time. Better on both numbers.

Slide 14 — Fairness

Time: ~1 min 30

- We did not use gender as a feature. But that on its own does not make a model fair, so we measured four things.
- **One** — there is a real gap in the bank's history. **54.1% of men** approved against **50.4% of women**. A 3.7-point gap.
- **Two** — but inside Gig/Contract that gap is **0.5 points**, and inside Business Owner it is **minus 0.6**, actually favouring women. So the gap is about **which groups women are in**, not about different treatment within them.
- **Three** — what explains it. Women in this file have a median income of **₹45,000 against ₹55,000**, and they are **almost twice as likely to be gig workers — 21% against 12%** — which is the lowest-approval category. Their credit scores are basically identical, 723 against 727.
- **Four** — our model does **not** widen it. At 0.90 it auto-declines **15.6% of women and 15.0% of men**, a 0.6-point gap against 3.7 in the human decisions.
- Our position: gender should **never** be a feature. But leaving it out is **necessary, not sufficient** — income and employment type are strong proxies, so a model can reproduce a gender gap without ever seeing the column. That is why we ask for monthly monitoring.

Slide 15 — What it is worth

Time: ~1 min 30

- So what is this actually worth to the bank? Five things.
- **One, capacity without hiring.** 190 hours a month is **2,280 hours a year — about 1.2 full-time analysts.** The unit can take roughly **15% more applications** with the team it already has.
- **Two, the time moves to where money is made.** Those hours shift from obvious no's onto **borderline files** — the ones where an approval is actually won or lost. A clear decline earns the bank nothing.
- **Three, good customers get answers faster**, because 456 files leave the queue and everyone behind them moves up. Fewer people give up and go to a competitor.
- **Four, one consistent written policy** — that 50% FOIR limit, applied the same way by every analyst.
- **Five, protecting tomorrow's customers** — the carve-out saves about **6 young first-time borrowers a month** from a wrong machine rejection. Those are the longest-lifetime customers a bank has.
- The yellow box puts a rupee figure on it, and I want to be honest about it. **The 2,280 hours is our number. The ₹500 an hour is our assumption** — Finance has to give the real rate. At that rate it is about **₹11.4 lakh a year**, before any revenue from faster decisions.

Slide 16 — What to do, in what order

Time: ~1 min

- Six actions, in order. Nothing here asks the bank to trust the model before it has watched it work.
- **One: shadow-run it for a month.** Score live applications, record what the model *would* have done, do not act on it. Compare against the analysts.
- **Two: launch at 0.90 with the carve-out**, then move to 0.88 once the shadow month confirms the rates.
- **Three: write the FOIR limit into policy. Four: start capturing repayment outcomes** — that is the single biggest upgrade available to the next version. **Five: monthly monitoring, including by gender. Six: recalibrate quarterly.**
- The ask is small and the exit is easy. A shadow month costs nothing and can be stopped at any point.

Slide 17 — Limitations and close

Time: ~1 min 15

- Finally, the four things we would raise ourselves if we were sitting on your side of the table.

- **The biggest one:** our target is a **human decision, not repayment**. When we say 4.7% wrong, we mean 4.7% **disagreed with an analyst** — not that 4.7% were good credit risks. The model has never seen whether a single customer repaid.
- Second, the threshold comes from 2,400 test applications, so expect real month-to-month movement. Third, our own cleaning choices move the recommended threshold between **0.85 and 0.90** — about 65 analyst hours a month. That is why we kept a cleaning log. Fourth, a model only knows the policy it learned; if the credit team changes how it lends, it has to be retrained.
- So, where we land: **auto-decline at 0.90, never auto-decline someone with no credit history, and shadow-run it for a month first.** That frees about **190 analyst hours a month** at a wrong-decline rate of **4.7%** — inside the limit the bank set itself.
- Thank you. We are happy to take questions.

Expected questions — and short answers

Answer in **two sentences**: the number first, then the reason. If you do not know, say "we did not test that" — that is a better answer than guessing.

Q. Why 0.90? Why not 0.85 or 0.95?

A. 0.90 is the **lowest** cut-off that keeps wrong declines under the bank's 5% limit — it is at 4.7%. 0.85 breaks the rule at 5.9%, and 0.95 is safe at 2.1% but removes only 10% of applications, so we would be paying in volume for safety the bank already said it does not need.

Q. Is 79% accuracy good enough?

A. Accuracy is the wrong question here. What matters is that at our threshold only **4.7% of auto-declined applications** were people the credit team would have approved — that is the number the bank set a limit on.

Q. What is a confusion matrix?

A. A 2-by-2 table of what the model predicted against what actually happened. Ours is **1,046 correctly approved, 852 correctly declined, 277 declines we missed, and 225 good customers wrongly declined** at the default cut-off.

Q. What is the difference between precision and recall?

A. **Precision** is about what the model says: of everyone it declines, how many really were declines — 79.1% for us. **Recall** is about reality: of all the real declines, how many it caught — 75.5%. Precision asks "was I right when I acted?", recall asks "did I catch everything?"

Q. Why predict Declined instead of Approved?

A. Because auto-declining is the **action** we are taking, so the class we care about should be the positive one. It makes the model's probability read directly as "how confident am I that this should be declined", which is what the threshold works on.

Q. Why did you keep the rows with no credit score?

A. Because they are not errors. **Credit history is zero months for all 1,039 of them** — they are new-to-credit applicants, 8.7% of all customers and mostly young. Dropping them would delete a real customer segment the bank lends to.

Q. Why fill missing income with the employment-type median and not the overall median?

A. Because income differs hugely by employment type — **₹28,000 for gig workers against ₹75,000 for self-employed professionals**. A single global median would overstate gig incomes and understate self-employed ones, and 86% of the missing values were self-employed, so we would have invented a bias ourselves.

Q. How do you know those 12 incomes were annual, not real?

A. Because dividing every one of them by 12 puts **all 12 back into a normal salaried band**, ₹51,500 to ₹1,38,000. A data-entry explanation that works for all 12 cases is far more likely than 12 genuine six-lakh monthly salaries.

Q. Why is loan-to-income negative in your model when bigger loans should be worse?

A. That is **collinearity**, not a finding, and we say so in the notebook. Loan-to-income correlates with FOIR and with income, so once FOIR is holding affordability constant, what is left picks up the fact that bigger loans go to higher earners. The honest relationship is in our Step 5.3 table: approval falls from 61.5% to 42.7%.

Q. Could you not just set a credit score floor of 650 and skip the model?

A. You could, and it would be almost as safe — it flags 9.8% of applications with 3.0% wrong. The model adds about **5 percentage points more volume**, roughly 160 applications and 67 analyst hours a month, from cases where FOIR and missed payments justify a decline that the score alone would not.

Q. Is the model fair?

A. It does not widen the historical gap — **15.6% of women auto-declined against 15.0% of men**, against a 3.7-point gap in the human decisions. But we are careful not to call it "fair": it copies past analyst decisions, so parity with the analysts is parity with a possibly biased baseline.

Q. You left gender out — doesn't that make it fair automatically?

A. No. **Income and employment type are strong proxies for gender** in this data, so a model can reproduce a gender gap without ever seeing the column. That is exactly why we recommend reporting outcomes by gender every month.

Q. What would make this model better?

A. **Repayment data.** Right now the target is what an analyst decided, not whether the customer repaid, so we are measuring agreement with ourselves rather than credit risk. For new-to-credit applicants specifically we would want bank-statement cash flow or rent and utility payment history.

Q. What happens to the 69% of declines the model does not catch?

A. They still go to an analyst, exactly as today. That is not a weakness — it is the price of the 5% rule, and the bank chose that limit. Once repayment data arrives, the limit can be revisited with evidence.

Q. Why not use a decision tree or random forest — wouldn't it be more accurate?

A. We have only covered logistic regression at this stage of the course, and the brief specifies it. It also happens to be the right tool here, because the credit team needs to read the coefficients and agree with them — a model they cannot explain is one they cannot defend to a regulator.

Q. How confident are you in the 4.7%?

A. Moderately. It comes from **2,400 test applications**, so it carries sampling error, and our own cleaning choices move the recommended threshold between 0.85 and 0.90. That is why we recommend a shadow month first and recalibrating quarterly.

Q. Who did what in the team?

A. Mutee did Step 5.1 cleaning and the CBIM canvas; Saalif did Steps 5.2 and 5.3, the drivers and the affordability features; Javeed did 5.4 and 5.5, the model and the threshold; Varshitha did 5.6 and 5.7, fairness and the briefing. The notebook was tested end-to-end by someone who did not write the section.

The ten numbers everyone must know

Whatever question you get, one of these is usually the answer. Learn these ten.

#	Number	What it is
1	**3,000 / 25 min / 1,250 hrs**	Applications a month, minutes per manual review, total analyst hours a month
2	**52.9%**	Overall approval rate — the target is close to balanced
3	**3.0% below 650 vs 32.5% above**	The credit score cliff — a tenfold jump at one point
4	**65% → 32% → 15.6%**	The FOIR cliff: under 30%, 50–55%, above 55%
5	**79.1% accuracy · 0.867 ROC-AUC**	Model performance on the 2,400-application test set
6	**225**	Good customers wrongly declined at the default 0.5 cut-off
7	**0.90 → 15.2% / 4.7%**	Our threshold: share auto-declined, share wrongly declined
8	**456 applications · 190 hours a month**	What the threshold is worth every month
9	**81.2% vs 56.7%**	Model accuracy with a credit score vs without one
10	**15.6% vs 15.0%**	Auto-decline rate for women vs men — the model does not widen the gap

If you are stuck mid-answer, fall back to this sentence: “The bank sets the risk limit, and the model's job is only to show where that limit falls.” It is true, it is the point of the whole use case, and it buys you ten seconds to think.